

AI-Driven Early Detection and Classification of Water Quality for Safe Drinking Water

TY B.Tech

**Computational Intelligence
Project Report**

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ABSTRACT

This project focuses on AI-driven early detection and classification of drinking water quality using machine learning techniques. The aim is to determine whether a given water sample is safe (potable) or unsafe (non-potable) for drinking based on its physical and chemical parameters such as pH, hardness, turbidity, and dissolved solids.

The dataset used for this project was obtained from the Kaggle Water Potability Dataset, containing detailed records of water samples. Data preprocessing steps included handling missing values, normalization, and feature scaling to ensure compatibility with various machine learning models.

Models such as Random Forest, AdaBoost, CatBoost, LightGBM, and XGBoost were implemented and evaluated using performance metrics like accuracy, precision, recall, and F1-score. Among these, CatBoost achieved the best accuracy of 84%, with high precision and recall values, making it the most effective model for classifying water safety.

This project also aligns with United Nations SDG 6 – Clean Water and Sanitation, promoting safe and sustainable access to potable water through data-driven insights and intelligent automation.

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1. INTRODUCTION

Access to safe drinking water is a global challenge, and contaminated water causes serious health risks. Traditional lab testing is accurate but slow and costly, whereas machine learning provides a faster and scalable solution.

The SafeSip project uses AI to predict water potability based on parameters like pH, hardness, turbidity, and conductivity, enabling smart and quick water quality assessment. The system includes data preprocessing, model training, evaluation, and a Streamlit-based SafeSip interface for real-time predictions.

1.1 OBJECTIVE

The primary objectives of this project are:

- To classify water samples as potable (safe) or non-potable (unsafe) using machine learning models.
- To assist in environmental and health safety monitoring.
- To promote early detection of water contamination and reduce health risks.
- To develop a web-based AI application for real-time prediction and visualization.

1.2 MOTIVATIONS

The motivation behind this project stems from the growing concern over water pollution and scarcity of safe drinking water. Manual testing methods are resource-intensive and not suitable for real-time detection.

AI and ML offer a new perspective to tackle this issue by leveraging data analytics for environmental monitoring. Accurate and quick classification of water quality can help authorities and households take timely action, preventing potential health hazards.

Furthermore, aligning the project with **SDG Goal 6: Clean Water and Sanitation**, the work emphasizes sustainable and intelligent solutions for global water safety.

2. METHODOLOGY

The project methodology integrates data preprocessing, model training, evaluation, and MLOps deployment for automation and monitoring.

1. Data Collection and Preprocessing

- Dataset Source: Kaggle Water Potability Dataset
- Features: pH, Hardness, Solids, Chloramines, Sulfate, Conductivity, Organic Carbon, Trihalomethanes, Turbidity
- Target Variable: Potability (1 = Safe, 0 = Unsafe)
- Preprocessing Steps:
 - Handling missing values using mean/median imputation
 - Normalization and scaling
 - Feature selection and correlation analysis

2. Exploratory Data Analysis

- Visualized feature correlations and probability distribution of potability.
- Dataset found to be fairly balanced ($\approx 50\%$ safe, 50% unsafe).
- Strong correlation observed between turbidity, pH, and hardness affecting water quality.

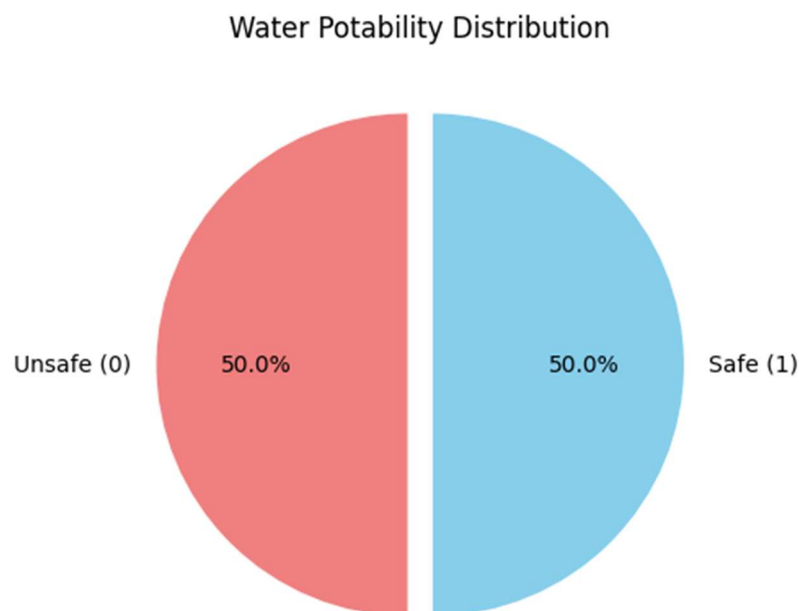


Fig 1 : Water Potability Distribution

3. ModelDevelopment

Trained and tested multiple algorithms:

- Random Forest
- AdaBoost
- CatBoost
- LightGBM
- XGBoost

Dataset split: 80% training, 20% testing Cross-validation used to ensure model reliability.

Confusion Matrix — Random Forest (Optuna Tuned)



Fig 2: Random Forest Confusion Matrix

Confusion Matrix — AdaBoost Model

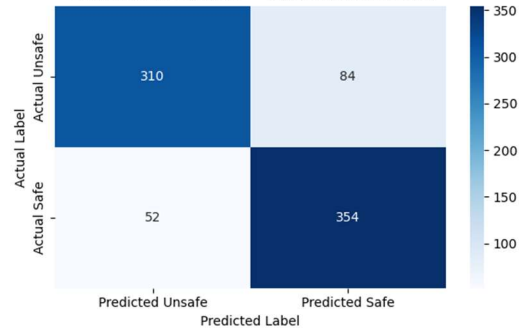


Fig 3 : AdaBoost model Confusion Matrix

Confusion Matrix — CatBoost Model

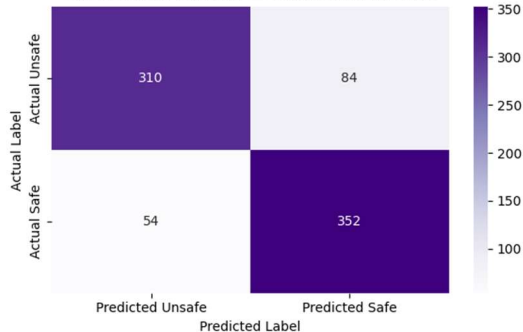


Fig 4 : CatBoost Model Confusion Matrix

Confusion Matrix — LGBM Model

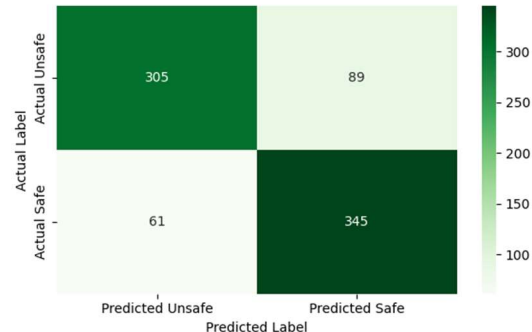


Fig 5 :LGBM Model Confusion Matrix

Confusion Matrix — XGBoost (Optuna Tuned)

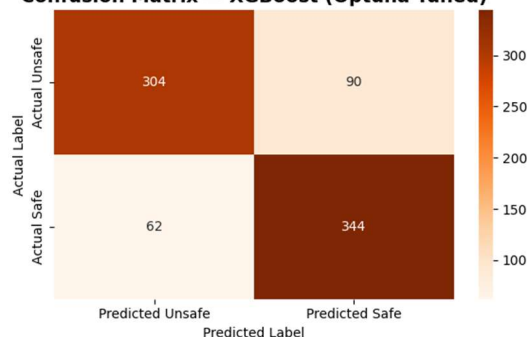


Fig 6 : XGBoost Confusion matrix

5. MLOps Workflow Integration

- Automated pipeline for:
 - Data preprocessing
 - Model training and evaluation
 - Deployment using Streamlit web app (SafeSip)
 - Monitoring and continuous improvement



Fig 7 : MLOps Workflow Integration

6. Deployment

The SafeSip web application enables users to:

- Enter physical and chemical water parameters
- Predict water safety instantly
- Upload CSV files for batch prediction
- View visual feedback and recommendations





SafeSip – Water Safety Checker



Ensuring pure & safe water for every life — powered by smart ML.

Enter Water Parameters

pH

Hardness

Solids

Chloramines

Sulfate

Conductivity

Organic Carbon

Trihalomethanes

Turbidity

Predict Potability

♥ The water is SAFE for Drinking — SafeSip indicates good potability.

Suggestion: Continue regular monitoring. Consider filtration if taste/odor issues are observed.

Fig 8. SafeSip Web Application Streamlit Deployment





Upload CSV for Batch Prediction

Upload CSV (columns: ph, Hardness, Solids, Chloramines, Sulfate, Conductivity, Organic_carbon, Trihalomethanes, Turbidity):



Drag and drop file here

Limit 200MB per file • CSV

Browse files



Upload_Example.csv 10.1 KB

✓ Batch predictions complete!

	ph	Hardness	Solids	Chloramines	Sulfate	Conductivity	Organic_carbon	Trihalomethanes	Turbidity	Prediction
0	-0.0263	0.2592	-0.1395	0.1124	0.9676	1.709	-1.1807	1.3055	-1.2863	Not Safe
1	-0.1864	1.3883	1.9538	1.5033	-4.0501	-0.6156	-0.8624	0.4675	-0.1433	Safe
2	0.6974	0.8477	-0.24	1.3606	-0.0043	-0.094	0.7811	0.0009	-1.1674	Not Safe
3	0.8455	0.5477	0.0005	0.592	0.6458	-0.7788	1.2551	2.1523	0.8484	Not Safe
4	-0.3546	0.0741	1.3679	0.9912	-1.6225	0.6306	0.0256	0.7463	0.9243	Safe
5	-1.0137	-0.2451	0.7682	0.267	-0.19	-1.8034	-1.7793	-0.7286	-1.8033	Not Safe
6	0.6021	-0.757	0.1838	1.526	-2.0355	-1.2199	0.5539	2.6224	0.9371	Safe
7	-2.1713	1.443	-1.6268	-4.1646	-0.0472	-1.0508	-0.3913	-0.3993	-0.2983	Safe
8	-0.0263	-2.3538	-0.8815	0.4308	-1.7957	-0.4557	-0.4774	-0.7913	-0.4765	Not Safe
9	2.7937	0.9388	0.3958	1.2351	1.9506	1.7037	1.1013	0.3533	0.5175	Not Safe

Fig 9. SafeSip Web Application Streamlit Deployment Batch Prediction

3. RESULTS

3.1 Model Performance

- CatBoost model achieved 84% accuracy, with high precision and recall balance.
- Demonstrated robustness on unseen data with minimal overfitting.

Model	Accuracy	Precision	Recall	F1-Score	R ² Score
Random Forest	0.83	0.812	0.865	0.838	0.32
AdaBoost	0.84	0.742	0.764	0.752	0.21
CatBoost	0.84	0.82	0.877	0.848	0.36
LightGBM	0.89	0.792	0.845	0.818	0.235
XGBoost	0.816	0.796	0.821	0.826	0.265

Fig 1 : Model Evaluation Table

CatBoost achieved the best balance of accuracy, precision, recall, and F1-score, making it ideal for deployment.

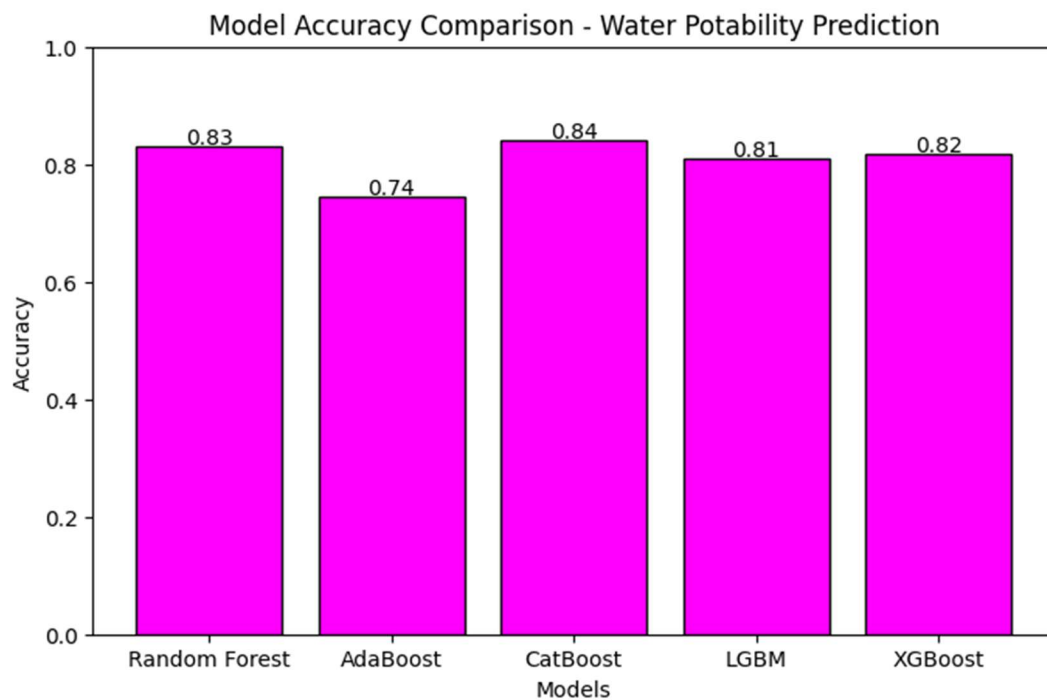


Fig 10 : Model Accuracy Comparison

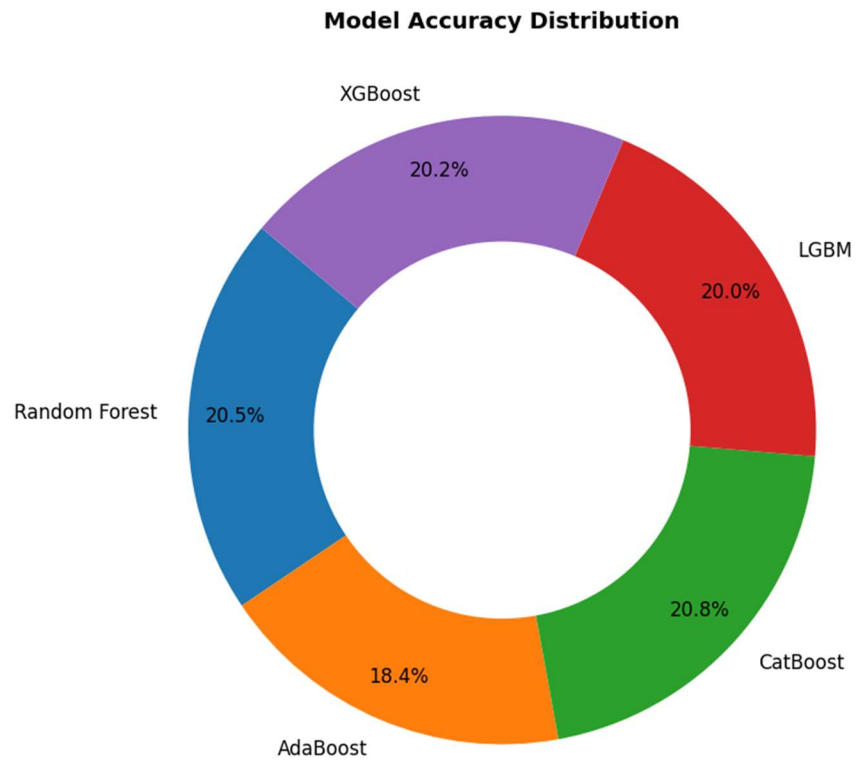


Fig 11 : Model Accuracy Distribution

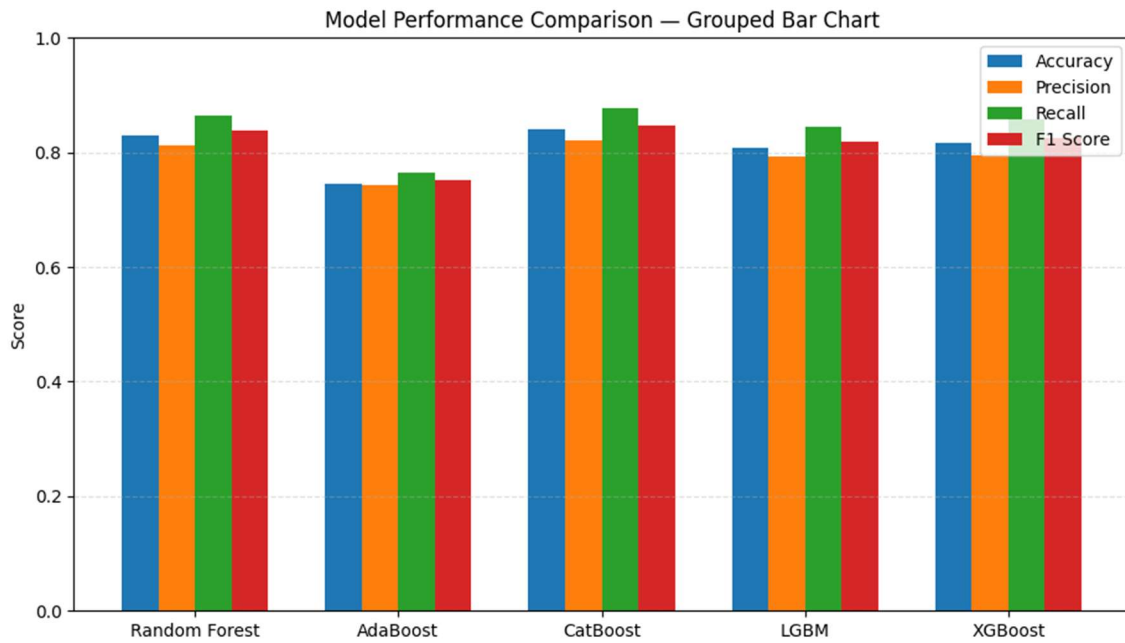


Fig 12 : Model Performance Comparison

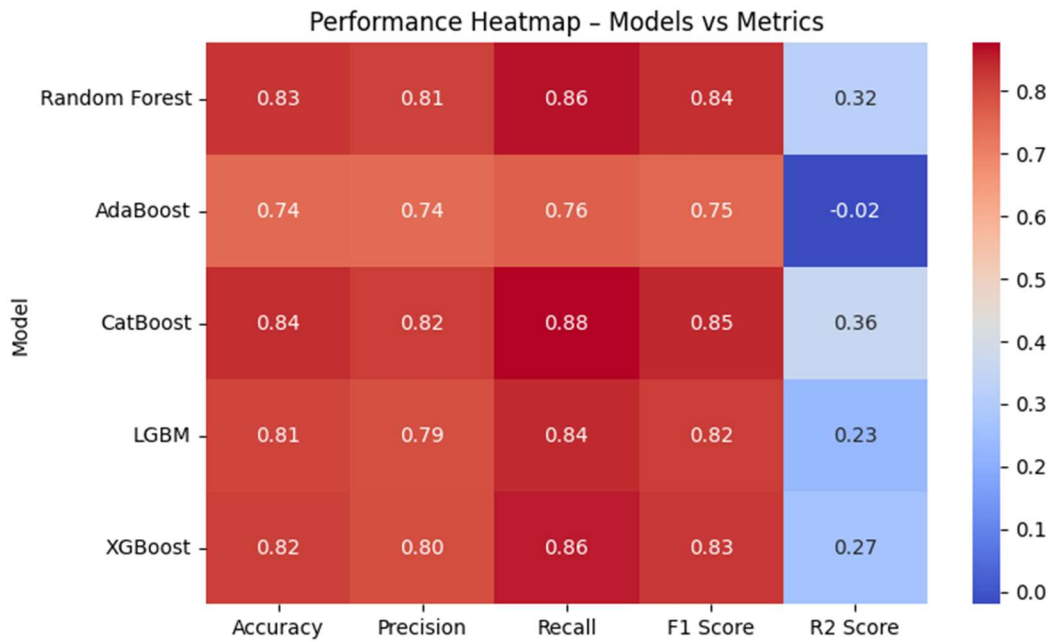


Fig 13 : Model Performance Heatmap

Key Insights

- High turbidity and hardness strongly indicate unsafe water.
- Neutral pH and moderate hardness values correspond to higher potability.
- Dataset analysis revealed potential for IoT sensor integration for real-time quality assessment.

Applications

- Smart water quality monitoring
- Health and safety surveillance systems
- Rural and urban water purification management
- Integration into IoT-based filtration systems

SDG Goal Alignment

- SDG 6: Clean Water and Sanitation
- Ensures sustainable and safe access to clean drinking water through AI-powered prediction.



Fig 14 : SDG 6 -Clean Water and Sanitation

Web App Outcome

- Real-time input-based prediction
- Batch CSV analysis for bulk data
- Suggestive outputs such as:
“The water is SAFE for Drinking — SafeSip indicates good potability.”

 The screenshot shows the 'SafeSip - Water Safety Checker' web application. The header includes the 'SafeSip' logo, the title 'SafeSip - Water Safety Checker', and the tagline 'Ensuring pure & safe water for every life — powered by smart ML.' Below the header is a search bar. The main section is titled 'Enter Water Parameters' and contains nine input fields for water quality parameters: pH (7.00), Hardness (180.00), Solids (15000.00), Chloramines (7.50), Sulfate (330.00), Conductivity (500.00), Organic Carbon (10.00), Trihalomethanes (70.00), and Turbidity (3.00). Each field has a slider control. A 'Predict Potability' button is located below the input fields. At the bottom, a purple banner displays the result: 'The water is SAFE for Drinking — SafeSip indicates good potability.' Below this banner is a suggestion: 'Suggestion: Continue regular monitoring. Consider filtration if taste/odor issues are observed.'

Fig 15 : Web-App Outcome

4 . CHALLENGES

Data Imbalance & Missing Values

- Addressed through imputation and normalization.

Overfitting

- Controlled using cross-validation and regularization.

Model Interpretability

- Improved using feature importance visualization and SHAP analysis.

Deployment Scalability

- Optimized Streamlit integration for lightweight web operation.

5. FUTURE IMPROVEMENTS

- Integration with IoT sensors for real-time water monitoring.
- Addition of deep learning models (ANN, CNN) for higher prediction accuracy.
- Development of mobile app interface for field testing.
- Advanced MLOps automation for continuous retraining.
- Inclusion of geographical and seasonal water quality analysis.

6. CONCLUSION

The SafeSip project successfully applies machine learning to predict and classify water quality as safe or unsafe for drinking. By implementing preprocessing, model comparison, and hyperparameter tuning, CatBoost emerged as the best-performing algorithm with strong generalization ability.

The deployment of the SafeSip web app demonstrates a practical and user-friendly approach to AI-based environmental monitoring. This project contributes to the advancement of smart water safety systems and aligns with SDG 6: Clean Water and Sanitation, ensuring safe water accessibility for all.

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